

Cyrus Guided Notes: Tsinghua Automation undergraduate program PDF

Tsinghua course pack

2026-05-15

Course source

Official source: <https://www.tsinghua.edu.cn/jxjyw/bkzy2023/zxzy/27.pdf>

Why this course is in the system. Undergraduate curriculum source for signals, automatic control, pattern recognition, machine learning, and robotics.

Learning output. Create a GoodNotes undergraduate spine page with prerequisites, formulas, and outputs.

Prerequisite checkpoint

- Advanced mathematics
- Linear algebra
- Programming

Formula checkpoint

Do not memorize these first. Read each formula as a sentence: what changes, what is observed, what is optimized, and what output it produces.

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau$$

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N \ell(f_{\theta}(\mathbf{x}_i), y_i)$$

$$\mathbf{s}\mathbf{u} = \mathbf{K}[\mathbf{R} \ \mathbf{t}]\mathbf{X}$$

Visual page

Turn the undergraduate program into a prerequisite graph instead of a calendar.

GoodNotes pages

- Page THU-B001: undergraduate course spine
- Page THU-B002: where each formula first appears

Web interaction

A filterable curriculum map by math, control, AI, robotics, and spatial intelligence.

Obsidian and Notion

Layer	Output
Obsidian	Tsinghua Automation -> undergraduate spine
Notion	Course-source evidence and next PDF expansion status.

First study move

Open the official source, copy the prerequisite checkpoint into GoodNotes, then write one formula in your own words before watching or reading more.